# A Literal Counterexample to the Printed Banach Projection Problem

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## Source Metadata

Source paper: K. Mahesh Krishna, *Paulsen and Projection Problems for Banach Spaces*, arXiv:2201.00991, 2022.

Target statement: Problem 2.14, "Projection problem for Banach spaces", on pages 7–8 of the source PDF.

## Open Problem Evidence

Here we state projection problem for Banach spaces.

**Problem 2.14.** ***(Projection problem for Banach spaces) Let*** $\mathcal{X}$ ***be a*** $d$***-dimensional Banach space with Auerbach basis*** $(\{\zeta_k\}_{k=1}^d, \{u_k\}_{k=1}^d)$ ***(***$\{\zeta_k\}_{k=1}^d$ ***is the collection of dual functionals).***

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***Find the function*** $g : (0,1) \times \mathbb{N} \times \mathbb{N} \to [0, \infty)$ ***satisfying the following: If*** $P : \mathcal{X} \to \mathcal{X}$ ***is a projection (idempotent) of rank*** $n$ ***satisfying***

$$(1-\varepsilon)\frac{n}{d} \leq \|Pu_k\|^2 = \|\zeta_k P\|^2 = |\zeta_k(Pu_k)| \leq (1+\varepsilon)\frac{n}{d}, \quad \forall 1 \leq k \leq d,$$

***then there exists a projection (idempotent)*** $Q : \mathcal{X} \to \mathcal{X}$ ***with***

$$\|Qu_k\|^2 = \|\zeta_k Q\|^2 = |\zeta_k(Pu_k)| = \frac{n}{d}, \quad \forall 1 \leq k \leq d,$$

***satisfying***

$$\sum_{k=1}^{d} \frac{1}{2}(\|Pu_k - Qu_k\|^2 + \|\zeta_k P - \zeta_k Q\|^2) \leq g(\varepsilon, n, d).$$

We already remarked that Paulsen and projection problems were connected using chordal distance. Even though we are unable to connect Paulsen and projection problems for Banach spaces, we introduce the

## Transcription of the Relevant Clause

Problem 2.14 asks for a function $g$ such that, whenever a rank $n$ projection $P$ on a $d$-dimensional Banach space with Auerbach basis $(\{\zeta_k\}_{k=1}^d, \{u_k\}_{k=1}^d)$ satisfies

$$(1-\varepsilon)\frac{n}{d} \leq \|Pu_k\|^2 = \|\zeta_k P\|^2 = |\zeta_k(Pu_k)| \leq (1+\varepsilon)\frac{n}{d} \quad (1 \leq k \leq d),$$

there is a rank $n$ projection $Q$ with

$$\|Qu_k\|^2 = \|\zeta_k Q\|^2 = |\zeta_k(Pu_k)| = \frac{n}{d} \quad (1 \leq k \leq d)$$

and a corresponding distance estimate.

## Idea of the Proof

The obstruction is contained in the displayed equality itself. The conclusion does not merely prescribe properties of the new projection $Q$; it also requires the old scalar $|\zeta_k(Pu_k)|$ to equal $n/d$. Hence any allowed $P$ whose displayed diagonal values are close to, but not exactly, $n/d$ immediately contradicts the conclusion. A two-dimensional Hilbert space already supplies such a $P$.

**Theorem 1.** *Problem 2.14 of arXiv:2201.00991 is false as printed.*

*Proof.* Let $X = \mathbb{R}^2$ with its Euclidean norm. Let $u_1 = e_1, u_2 = e_2$, and let $\zeta_1, \zeta_2$ be the coordinate functionals. This is an Auerbach basis. Put $d = 2$, $n = 1$, and $\varepsilon = 1/2$, so that

$$\left[(1-\varepsilon)\frac{n}{d}, (1+\varepsilon)\frac{n}{d}\right] = \left[\frac{1}{4}, \frac{3}{4}\right].$$

Let

$$v = \sqrt{\frac{3}{5}}\, e_1 + \sqrt{\frac{2}{5}}\, e_2$$

and let $P$ be the orthogonal projection onto $\mathbb{R}v$:

$$Px = \langle x, v\rangle v.$$

Then $P$ is an idempotent rank-one projection. Since $v$ is a unit vector,

$$Pe_1 = \sqrt{\frac{3}{5}}\, v, \qquad Pe_2 = \sqrt{\frac{2}{5}}\, v.$$

Therefore

$$\|Pe_1\|^2 = \frac{3}{5}, \qquad \|Pe_2\|^2 = \frac{2}{5}.$$

Moreover,

$$\zeta_1 P : x \mapsto \sqrt{\frac{3}{5}}\, \langle x, v\rangle, \qquad \zeta_2 P : x \mapsto \sqrt{\frac{2}{5}}\, \langle x, v\rangle,$$

so

$$\|\zeta_1 P\|^2 = \frac{3}{5}, \qquad \|\zeta_2 P\|^2 = \frac{2}{5}.$$

Finally,

$$|\zeta_1(Pe_1)| = \frac{3}{5}, \qquad |\zeta_2(Pe_2)| = \frac{2}{5}.$$

Both $\frac{3}{5}$ and $\frac{2}{5}$ lie in $[\frac{1}{4}, \frac{3}{4}]$, so $P$ satisfies the hypothesis of Problem 2.14.

However, the printed conclusion requires, for every $k = 1, 2$,

$$\|Qu_k\|^2 = \|\zeta_k Q\|^2 = |\zeta_k(Pu_k)| = \frac{n}{d} = \frac{1}{2}.$$

For $k = 1$ this asserts $\frac{3}{5} = \frac{1}{2}$, and for $k = 2$ it asserts $\frac{2}{5} = \frac{1}{2}$. These contradictions are independent of the choice of $Q$. Thus no projection $Q$ can satisfy the printed conclusion, and no finite function $g$ can make Problem 2.14 true as stated. □

## Verification Notes

The arithmetic is exact. The included script `code/verify_counterexample.py` checks that the two diagonal values 3/5 and 2/5 lie in the required $\varepsilon = 1/2$ interval and are not equal to 1/2.

Command used:

```
python code/verify_counterexample.py
```

The result is a literal counterexample to the source statement as printed. It is not a counterexample to a corrected version in which the conclusion asks for $|\zeta_k(Qu_k)| = n/d$ instead of $|\zeta_k(Pu_k)| = n/d$.

## Novelty and Search Bounds

The cheap run indexes were searched for arXiv:2201.00991, Paulsen, projection problem, approximate Schauder frame, and related phrases. No prior packet for this source was found. Web searches on 2026-06-28 for the exact title and projection-problem phrases returned the source arXiv record and no separate correction or later answer. This novelty check is therefore bounded; the claim should be reviewed as an erratum-style literal counterexample.

## Human Review Recommendation

Review the exact printed equality in Problem 2.14. If it is a typographical error and the intended equality contains $Q$ rather than $P$, this packet should be recorded as a formulation issue only. The corrected Banach projection problem is not addressed here.

## References

[1] K. Mahesh Krishna. *Paulsen and Projection Problems for Banach Spaces*. arXiv:2201.00991, 2022.